

• **ADFORGE AI — PROJECT DOCUMENTATION**

• **Solution Architecture Definition**

- **Date:** 7th July 2026
- **Development Profile:** Sunehri Sharma
- **Maximum Marks:** 5 Marks

Proposed Architecture Diagram Components

Here is how your local Flask application layers align with the template's structural boxes:

- **Presentation / Client Layer:** Dark-Themed Glassmorphism Workspace Dashboard (HTML5, CSS3, Vanilla JavaScript with Fetch API). Handles input collection forms, asynchronous generation triggers, and output copy canvas blocks.
- **Auth & Configuration Service:** Local System Environment Variable Controller (python-dotenv checking .env variables for secure API configuration storage).
- **API Gateway / Server Router:** Flask Micro-Application Server (app.py running on localhost:5000 or exposed via public pyngrok secure endpoints).
- **Core Logic Service:** Prompt Orchestration Engine & Asynchronous Change Handlers. Packages user descriptions into valid system instructions and processes external responses.
- **External API Integrations:** Groq Cloud API Gateway executing ultra-low latency requests to the LLaMA 3.3-70B-Versatile model.
- **Data / Storage Layer:** Zero-Persistence In-Memory Cache (Local browser session variables and active server memory cache handling prompt revision logs).

Proposed Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Renders the single-page creative dashboard workspace. It collects custom text parameters (USPs, names, voices), triggers non-blocking execution requests, manages responsive grid panels, and provides quick clipboard copy controls.	HTML5, CSS3 (Custom Glassmorphic styles), Vanilla JavaScript (Fetch API).
[API Gateway /	Serves as the central backend controller. It intercepts inbound traffic request parameters from	Python / Flask Framework, pyngrok

Component Name	Description / Role in Architecture	Technologies Used
Router]	the user interface, manages application endpoints (such as /api/generate), applies security boundaries to key environments, and routes payload files.	(for secure public delivery tunnels).
[Core Microservices]	The foundational optimization block. It aggregates raw text forms into precise prompts, specifies structure profiles (like Google's strict 30-character constraint), runs the interactive "Refine Copy" loops, and handles external text synthesis pipelines.	Python Prompt Processing Scripts, Groq SDK Orchestrator (LLaMA 3.3-70B).
[Database / Storage Layer]	Provides a secure execution sandbox for short-term data management. By keeping transient parameters stored directly inside live local system memory or runtime cache buffers, it protects brand data privacy without writing briefs permanently to a hard drive.	System Environment Memory, Local Browser Session Storage.